

Networks: flows

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Solution of the practice quiz

There are three simple paths from v_0 to v_5 .

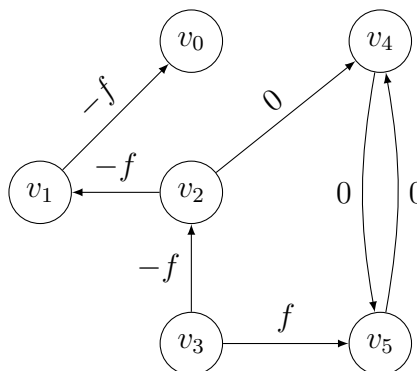
- The first one corresponds to

$$\mathcal{P}_1 = v_0 \leftarrow v_1 \leftarrow v_2 \leftarrow v_3 \rightarrow v_5.$$

Remember that if a flow f follows a simple path $\mathcal{P}$ from its origin to its destination, then we have

$$x_{ij} = \begin{cases} f & \text{if } (i, j) \in \mathcal{P}^{\rightarrow}, \\ -f & \text{if } (i, j) \in \mathcal{P}^{\leftarrow}, \\ 0 & \text{otherwise.} \end{cases}$$

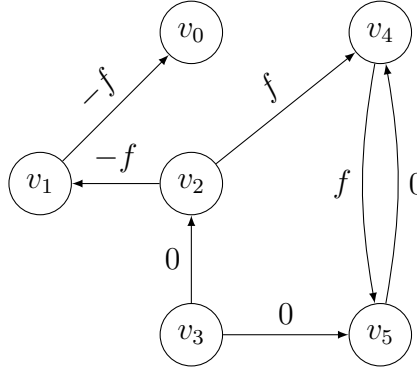
Here we have $\mathcal{P}_1^{\rightarrow} : \{(v_3, v_5)\}$ and $\mathcal{P}_1^{\leftarrow} = \{(v_1, v_0), (v_2, v_1), (v_3, v_2)\}$. Therefore the simple path flow for $\mathcal{P}_1$ is:



- The second one corresponds to

$$\mathcal{P}_2 : v_0 \leftarrow v_1 \leftarrow v_2 \rightarrow v_4 \rightarrow v_5.$$

Here we have $\mathcal{P}_2^{\rightarrow} : \{(v_2, v_4), (v_4, v_5)\}$ and $\mathcal{P}_2^{\leftarrow} = \{(v_1, v_0), (v_2, v_1)\}$.
Therefore the simple path flow for $\mathcal{P}_2$ is:



- The third one corresponds to

$$\mathcal{P}_3 = v_0 \leftarrow v_1 \leftarrow v_2 \rightarrow v_4 \leftarrow v_5.$$

Here we have $\mathcal{P}_3^{\rightarrow} = \{(v_2, v_4)\}$ and $\mathcal{P}_3^{\leftarrow} = \{(v_1, v_0), (v_2, v_1), (v_5, v_4)\}$.
Therefore the simple path flow for $\mathcal{P}_3$ is:

